1. Find the maximum (M) and minimum (m) values of $f(x,y) = 2x + 6y$ subject to the constraint $x^2 + y^2 = 10$.

Proof. Let $g(x,y) = x^2 + y^2 - 10$. We want points satisfying $\nabla f = \lambda \nabla g$ and $g(x,y) = 0$. We compute $\nabla f(x,y) = \langle 2, 6\rangle$ and $\nabla g(x,y) = \langle 2x, 2y\rangle$. Thus, we must solve

$$\begin{cases} 2 = 2\lambda x, \\ 6 = 2\lambda y, \\ x^2 + y^2 = 10. \end{cases}$$

The first two equations reduce to $\lambda x = 1$ and $\lambda y = 3$. Since $\lambda x = 1$, we know $\lambda \neq 0$ and $x \neq 0$. Dividing the second equation by the first gives

$$\frac{\lambda y}{\lambda x} = 3 \qquad \Longrightarrow \qquad y = 3x.$$

Substituting into the constraint, we obtain

$$x^2 + (3x)^2 = 10 \qquad \Longrightarrow \qquad 10x^2 = 10 \qquad \Longrightarrow \qquad x = \pm 1.$$

Hence, the candidate points are $(1,3)$ and $(-1,-3)$. Evaluating f at these points gives

$$f(1,3) = 2(1) + 6(3) = 20 \qquad \text{and} \qquad f(-1,-3) = 2(-1) + 6(-3) = -20.$$

Since the constraint is a circle, which is closed and bounded, these are the absolute maximum and minimum values. Therefore, $M = 20$ and $m = -20$.

Note A. The equation $\lambda x = 1$ does *not* imply $\lambda = 1$ or $x = 1$. For example, $\lambda = 2$ and $x = \frac{1}{2}$ also satisfy this equation. We should not split into the cases $\lambda = 1$ and $x = 1$. Instead, we use $\lambda x = 1$ and $\lambda y = 3$ together to obtain $y = 3x$, and then use the constraint to find the possible points. □

2. Evaluate $I = \int_0^{\sqrt{2\pi}} \int_{y^2}^{2\pi} y\cos(x^2)\,dx\,dy$ by switching the order of integration.

Proof. The original bounds describe the region

$$R = \{(x,y) : 0 \le y \le \sqrt{2\pi},\ y^2 \le x \le 2\pi\}.$$

This region is bounded by $x = y^2$, $y = 0$, and $x = 2\pi$. Reversing the order, we instead have $0 \le x \le 2\pi$ and $0 \le y \le \sqrt{x}$. Therefore,

$$I = \int_0^{2\pi} \int_0^{\sqrt{x}} y\cos(x^2)dydx = \int_0^{2\pi} \left[\frac{y^2}{2}\cos(x^2)\right]_{y=0}^{y=\sqrt{x}} dx = \frac{1}{2}\int_0^{2\pi} x\cos(x^2)dx = \frac{1}{4}\left[\sin(x^2)\right]_0^{2\pi}$$
$$= \frac{\sin(4\pi^2)}{4}.$$

□